Database Systems viva (ER modelling): student scripts

DS1030 | one script per student | 1 to 1 minute 30 seconds

How the recording works

- The examiner asks the question below, then you answer in your own voice using your script. You may glance at the script, but try to say it naturally.
- Treat it like a short presentation: greet the audience, talk to the camera, and say thank you at the end. The greeting and the thank-you are part of your script.
- Sit in front of the camera with your face clearly visible and nobody else in the picture. Do not draw anything.
- Practise your script once or twice first. Aim for the time written on your page.
- Do exactly the style described on your page. Each student is deliberately different, so please do not copy another student.
- Keep to your script. Do not add extra facts that are not in it.

The question asked to every student

> *A university has students, courses and lecturers. Students enrol in courses, and each course is taught by one lecturer. Without drawing, tell me the entities and their attributes, the relationships and their cardinality, the keys you would use, and give one example of a weak entity.*

Student 01

Complete and fluent

Your style: The strongest answer. Calm, clear, steady pace.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:14.

How to deliver it

- Speak at a normal, steady pace.
- Look at the camera, keep a relaxed expression, no fillers.
- Say every sentence in full.

Your script

Good morning, everyone, and thank you for listening. Today I would like to explain entity-relationship modelling, using the university example. Let me start with the entities. An entity is a real-world object we store data about, so here we have Student, Course and Lecturer, and an entity set is the collection of all entities of one type, for example all the students. Each entity is described by attributes. A student has a simple attribute like a student ID, a composite attribute like a full name made of first and last name, a multivalued attribute such as several phone numbers, and a derived attribute like age, which we work out from the date of birth. Moving on, a relationship connects entities: a student enrols in a course, and a lecturer teaches a course. For cardinality, a lecturer teaches many courses, so that is one-to-many, students and courses are many-to-many, and a lecturer with one office is one-to-one. Next, the primary key identifies each entity, like the student ID, and a foreign key points to another table, like the lecturer ID stored in the course table. Finally, a weak entity depends on another entity, for example a lecturer's dependent, identified together with the lecturer ID. Thank you very much.

Student 02

Strong content, but rushed

Your style: Knows everything, but talks far too fast and runs sentences together.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:04.

How to deliver it

- Speak fast, almost without breathing, but still clearly.
- Keep your face on camera. Hardly any pauses.
- Do not add fillers. The problem is speed only.

Your script

Hi everyone, thanks for having me, and I will be quick because there is a lot to cover. So entity-relationship modelling for the university, entities are the real-world objects we store data about, Student, Course and Lecturer, and an entity set is all the entities of one type, like all the students, and each one has attributes, which are the details we record about it, like a student ID, a composite attribute like a full name split into first and last name, a multivalued attribute like several phone numbers, and a derived attribute like age which is worked out from the date of birth, and then a relationship links entities, a student enrols in a course and a lecturer teaches a course, and the cardinality says how many on each side, a lecturer teaches many courses so that is one-to-many, and students and courses are many-to-many, and then the primary key is what identifies each entity, the student ID, and a foreign key is a column that points to another table, like the lecturer ID in the course table, and a weak entity cannot be identified by itself, like a lecturer's dependent, so that is basically everything we need for the university database, and I could keep going, but I hope that makes sense, thank you very much.

Student 03

Hesitant and uncertain

Your style: Mostly correct, but full of hedges and fillers.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:25.

How to deliver it

- Say the "um" and "uh" where they are written, as people naturally do, and let a short silence happen now and then.
- Sound unsure. Look at the camera, but let your face look worried.

Your script

Um, hello everyone, and thank you for listening. So, uh, today I am going to talk about the university database, which has students, courses and lecturers, and how we plan it before we build it. So, first, an entity is, um, a real-world thing that we keep data about, like a student, a course or a lecturer. And an entity set is, uh, all the entities of the same type, like all the students together. Then attributes are, um, the details about each entity, like the name, the age and the student ID. I think the next part is the relationships. A relationship is, uh, how two entities are linked, so a student enrols in a course, and a lecturer teaches a course. For the cardinality, um, that is how many are on each side. One lecturer teaches many courses, so that is one-to-many, and students and courses are many-to-many, because one student takes many courses and one course has many students. Then, uh, the primary key is the attribute that identifies each entity in a unique way, like the student ID, so no two students share it. And a foreign key is, um, a column that points to the primary key of another table, so we can join them. So, um, that is what I wanted to say. Okay, thank you very much.

Student 04

Entities and attributes only

Your style: Answers the first half of the question well and skips the rest.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:22.

How to deliver it

- Steady, flat, serious voice.
- Stop after derived attributes as if that was the whole answer. Do not mention relationships, keys or weak entities.

Your script

Hello everyone, and thank you for coming. Today I want to explain the first part of database design, which is entities and their attributes, and I will keep it simple, because this is the best place to start. First of all, an entity is a real-world object or thing that we want to store data about. In a university, the entities are the student, the course and the lecturer. An entity set is the collection of all the entities of the same type, so all the students together form one entity set, and all the courses form another. Second, each entity is described by its attributes, which are the properties of that entity. A student has attributes like the name, the date of birth and the phone number. There are different kinds of attributes. A composite attribute can be divided into smaller parts, for example a full name that is made of a first name and a last name. A multivalued attribute can hold more than one value for one entity, for example a student who has two or three phone numbers. And a derived attribute is not stored, because it can be worked out from another attribute, for example the age, which comes from the date of birth. These are the building blocks of every design, and we need to choose them carefully. That is what I wanted to share today. Thank you for listening, and I hope it was useful.

Student 05

Confident but wrong

Your style: Sounds sure of himself or herself, but nearly every definition is wrong.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:20.

How to deliver it

- Bright, confident, energetic voice. Smile.
- Say the wrong facts exactly as written, with total confidence.

Your script

Good afternoon, everybody. I am very happy to be here, because this is a topic I know really well, and I have explained it to many friends. Let me begin. Please listen carefully, because this is very important. An entity is just one piece of data, like a name or a phone number, nothing more than that. An attribute is a table, and every table must have at least ten attributes. A composite attribute means an attribute that belongs to two tables at the same time, and a multivalued attribute is an attribute that is never allowed to change after it is saved. Moving on, a relationship is the drawing of the boxes, so it is done by the artist, and cardinality is the total number of rows in the database. A one-to-one relationship means that everyone in the table has the same value, and a many-to-many relationship means the tables are too big and have to be deleted. The primary key is the attribute that is allowed to repeat as many times as you like, because it is the most common value. Finally, a weak entity is an entity that is not important, so we can remove it from the design, and weak entities are usually deleted at the end of each year. I am completely sure about all of this, because we always do it this way in my team, and if anyone disagrees with me, they are simply wrong, because I have checked it many times. So that is the full answer, and thank you very much for listening.

Student 06

Slow, quiet and trailing off

Your style: Soft, slow voice, long gaps, sentences that fade away.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:21.

How to deliver it

- Speak quietly and slowly, about half your normal speed.
- Let your voice fade a little at the end of each sentence, and leave a natural gap before the next one.
- Keep your face on camera.

Your script

Hello, everyone. Thank you for letting me speak. Please excuse my voice, it is a little quiet today. Today I will talk a little about database design, and I will try to explain it slowly. An entity is something we keep data about, like a student, or a course, or a lecturer. In a university, we always keep data about the student first. Attributes are the details of the entity, like the name of the student, and the student number. Next, a relationship is how the entities are connected, for example a student enrols in a course, and we show it with a line. The primary key is the attribute that identifies each one in a unique way, so that we can find the right student, and without it we cannot find anyone in the list. And a foreign key is, I think, the column that points to the key of another table, so that the tables are connected. This is important, because without keys the data becomes confusing very quickly. I learned this in the first week, and the lecturer said it many times. That is what I remember. I know there are some more ideas, but I cannot say them clearly now. I am sorry. Thank you for listening, and thank you for your patience.

Student 07

Explains with an example

Your style: Warm, expressive storyteller who explains everything through one student, Nimal.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:25.

How to deliver it

- Natural, friendly voice, smile, and use small hand gestures if you like.

Your script

Hello everyone, it is lovely to be here. Today I would like to explain database design, and I will use one student as my example, Nimal, because it makes everything easy to picture. Nimal is an entity, a real-world thing we store data about, and so are his courses and his lecturers. Nimal has attributes, like his student ID and his email. His full name is a composite attribute, because it splits into first name and last name. He gave two phone numbers, so that is a multivalued attribute, and his age is a derived attribute, because we work it out from his date of birth. Now, Nimal enrols in courses, and that is a relationship. Cardinality tells us how many on each side: Dr Perera teaches Databases and Networks, so one lecturer teaches many courses, which is one-to-many, and Nimal takes many courses while each course has many students, which is many-to-many. The student ID is the primary key, so no other student has it. The course table keeps the lecturer ID as a foreign key, so when we look at a course, we can find its lecturer quickly. To finish, a weak entity is like a lecturer's child in the staff records. The child has a name, but the name alone is not enough, because two lecturers could have children with the same name, so we need the lecturer's ID as well. That is how one student shows us the whole design. Thank you, I hope that was clear.

Student 08

Very short and vague

Your style: Barely remembers anything and says so.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:27.

How to deliver it

- Slow and unsure. Take your time between sentences.
- Look at the camera, but look a little lost.

Your script

Hello everyone. Um, thank you for giving me the chance to speak. I am a little nervous, because I did not sleep well last night, but I will try my best. So, today the topic is database design. A database is important, because every university has to keep the data of its students, the courses and the lecturers somewhere safe. If the data is wrong, the students will not be happy and the university will lose its good name, and sometimes people lose their jobs. So the design should be done carefully, and not in a hurry at the end of the project. My friend told me that a good design saves a lot of time later, and I believe him. There are different parts in the design. There is the entity, that is something we want to keep data about, like a student. And there are attributes, that is the details of it, like the name, the address or the age. And the primary key, I think that is the special attribute that is different for every student, like the student number, so that two students never get the same one. I do not really remember the other parts, we learned them last semester, and the lecturer showed us a long diagram. I know they are all important, but I would need to look at my notes. Maybe next time I can explain them better, because I know I did not answer everything today. Sorry, that is all I can say. Thank you for your time, and thank you for listening, everyone.

Student 09

Reads a flat list

Your style: Reads one-line definitions in a flat voice, with no explanation.

Aim for between 1:00 and 1:30. Most people finish this script in about 1:27.

How to deliver it

- Read it from your phone propped at camera level, so your face stays towards the camera.
- Flat, monotone voice, no smile. Leave a small gap after each definition.

Your script

Good morning, everyone. Thank you for coming. This is my presentation on database design. The topic of my presentation is the main terms. I will list the terms one by one, and I will read the definition of each one, and one example for each one. The first term is entity: a real-world object that we store data about. For example, a student. The second term is entity set: all the entities of one type. For example, all the students. The third term is attribute: a property of an entity. For example, the name of the student. The fourth term is relationship: a link between entities. For example, a student enrols in a course. The fifth term is cardinality: how many entities are on each side of a relationship. The sixth term is primary key: an attribute that identifies each entity. For example, the student ID. The seventh term is foreign key: an attribute that refers to the primary key of another table. For example, the lecturer ID in the course table. The eighth term is weak entity: an entity that depends on another entity. For example, the dependent of a lecturer. Those are the terms that I was asked about. I have listed all of them in the order that I learned them, and I have read the definition of every term. That is my answer, and that is the end of my presentation. Thank you very much for your time and for your attention. If you have any questions, please ask me now.